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Week 3: Exploratory Data Analysis

AI/ML in Biomarker Discovery — miRNA in Alzheimer’s Disease

Learning Objectives

By the end of Week 3, you will be able to: 1. Compute and interpret descriptive statistics for miRNA expression data, including coefficient of variation, detection (floor values on arrays, zero-inflation in counts), and explain what these statistics reveal about data quality and biological signal 2. Perform Principal Component Analysis on a miRNA expression matrix, interpret scree plots and loadings, and explain in biological terms what each principal component captures 3. Execute hierarchical and k-means clustering, evaluate optimal cluster number using gap statistic and silhouette width, and compute cluster purity against known disease groups 4. Produce and read a publication-quality annotated heatmap of miRNA expression, interpreting row and column structure in terms of disease biology 5. Identify and quantify the contribution of biological confounders (age, sex) to principal components using correlation tests and partial R^2 , and describe a strategy to account for these in downstream ML modeling

Conceptual Overview: Why Explore Before You Model?

When a clinician sees a new patient for the first time, they do not immediately prescribe a treatment. They take a history, examine the patient, order basic tests, and build up a picture before making decisions. The same principle applies in machine learning.

Exploratory Data Analysis (EDA) is the systematic examination of a dataset — its structure, its variation, its distributions, its outliers, and its relationships — before any supervised modeling. In the context of our miRNA Alzheimer’s disease dataset, EDA serves four essential purposes:

1. Catching problems that QC missed. Week 2 removed technically failed samples. EDA catches subtler issues: miRNAs that sit at the detection floor in most samples, outlier samples that passed QC metrics but sit far from all biological clusters in PCA space, or confounders (age, sex) that account for more variance than disease group.

2. Building biological intuition. Before you ask a machine learning model “which miRNAs discriminate AD from controls?”, you should have a personal sense of the data. Which miRNAs are most variable? Do the AD and control samples look visually separable? Which samples are the most unusual? EDA builds the biological intuition that lets you critically evaluate ML outputs rather than passively accept them.

3. Informing modeling decisions. The dimensionality of your data ($\sim 1,330$ samples $\times$ ~ 700 -900 miRNAs), the degree of class separation visible in PCA, the presence of confounders — all of these findings from EDA directly determine choices in Week 4: how many features to carry into ML, whether to include age/sex as covariates, which cross-validation strategy to use.

4. Generating preliminary biological hypotheses. A miRNA that appears in the top cluster-defining rows of a heatmap, and whose cluster separates AD from control perfectly, is a candidate worth investigating mechanistically — even before the formal differential expression analysis.

The biology-first principle applies here more than anywhere else in the course.

Every plot you generate this week should prompt a biological question. A cluster in a heatmap or PCA plot is not just a visual grouping; it represents a set of patients whose miRNA profiles are similar — and asking *why* they are similar is the beginning of a discovery.

MODULE 3.1 — Descriptive Statistics for Expression Data

3.1.1 What Are We Measuring?

Before applying any sophisticated algorithm, we characterize the basic statistical properties of our miRNA expression matrix. Our working matrix, loaded from Week 2 outputs, has the following

structure:

Rows = miRNAs (e.g., hsa-miR-21-5p, hsa-let-7a-5p, ...) – 925 after Week 2 filtering (694 after Week 1 filtering)
Columns = Samples (e.g., GSM3403761, GSM3403762, ...) – 1,328 samples (1,009 AD, 32 MCI, 29 NC)
Values = normalized log2 microarray intensity (GSE120584) – continuous, ranging ~ -1 to 16

For each miRNA (each row), we compute a set of statistics that describes its behavior across all samples.

3.1.2 Per-miRNA Descriptive Statistics

Mean expression: The average expression of a miRNA across all samples. Biologically, a miRNA with very low mean expression may not be reliably detected in serum and may be contributing more noise than signal.

Standard deviation (SD): Measures spread around the mean. A miRNA with high SD is more variable across samples — potentially because it changes between disease groups (interesting) or because it is noisy (less interesting).

Coefficient of Variation (CV): $CV = SD / \text{mean} \times 100\%$. CV expresses variability *relative to the mean*, enabling comparison of variability across miRNAs at very different expression levels. A miRNA with mean expression of 10 and SD of 2 has a CV of 20%; another with mean of 3 and SD of 1 has a CV of 33%. Without the CV, the second miRNA would appear less variable (SD of 1 < SD of 2), but relative to its expression level it is actually more variable.

Interquartile Range (IQR): The range from the 25th to the 75th percentile. IQR is more robust than SD to outlier samples. We use IQR-based variance filtering (keep top 75% by IQR) before most analyses.

Percentage of zeros (% zeros): In RNA-seq data, miRNAs with many zero counts — even after normalization — were not detected in many samples. A miRNA with 80% zeros is poorly expressed and will not contribute meaningful biological signal. **On a microarray such as GSE120584 this column is always 0:** undetected miRNAs are reported at a “not detected” floor value instead of zero (Week 2, Module 2.3.1), and Week 2 already removed miRNAs that sit at the floor in most samples.

Percentage at the detection floor (% at floor): The microarray equivalent of % zeros. In each sample the lowest value is the “not detected” floor (the same definition Week 2 used to count detected miRNAs), so % at floor is the share of samples in which the miRNA was not detected. In GSE120584 it ranges from 0% to 43%.

CV on a log2 scale: $CV = SD / \text{mean}$ assumes a ratio scale with a true zero. On log2 values the mean depends on an arbitrary origin, and miRNAs near the detection floor have small means — so they get inflated CVs simply because their mean is small. In GSE120584 the top-CV miRNAs (e.g. hsa-miR-3194-3p, hsa-miR-223-3p) all have low mean expression and sit at the detection floor in 9–43% of samples, while the most stable ones ($CV < 2\%$) are highly expressed and never at the floor. One of them, miR-4463, is an internal-control miRNA the submitters normalised to. Use CV as a rough screen only.

```
# =====  
# Load clean data from Week 2  
# =====  
expr <- readRDS("data/processed/GSE120584_expr_clean.rds")  
meta <- readRDS("data/processed/GSE120584_metadata_clean.rds")  
  
GROUP_COLOURS <- c(  
  "Control" = "#4575B4",  
  "Mild Cognitive Impairment" = "#FEE090",  
  "Alzheimer's Disease" = "#D73027"
```

```

)

# =====
# Compute per-miRNA descriptive statistics
# =====
# Detection floor (as in Week 2): each sample's lowest value = "not detected"
sample_floor <- apply(expr, 2, min)
at_floor      <- sweep(expr, 2, sample_floor + 0.01, "<=")

mirna_stats <- data.frame(
  mirna      = rownames(expr),
  mean_expr  = rowMeans(expr),
  sd_expr    = apply(expr, 1, sd),
  median_expr = apply(expr, 1, median),
  iqr_expr   = apply(expr, 1, IQR),
  pct_zeros  = rowSums(expr == 0) / ncol(expr) * 100,
  pct_at_floor = rowMeans(at_floor) * 100,
  stringsAsFactors = FALSE
)

mirna_stats$cv <- mirna_stats$sd_expr / mirna_stats$mean_expr * 100

# Sort by CV (most variable first)
mirna_stats <- mirna_stats[order(mirna_stats$cv, decreasing = TRUE), ]

# Preview most variable miRNAs
cat("=== Top 10 Most Variable miRNAs (by CV) ===\n")
print(head(mirna_stats[, c("mirna", "mean_expr", "sd_expr", "cv", "pct_zeros", "pct_at_floor")], ))

# Preview most stable miRNAs (lowest CV)
cat("\n=== Top 10 Most Stable miRNAs (by CV) ===\n")
print(tail(mirna_stats[, c("mirna", "mean_expr", "sd_expr", "cv", "pct_zeros", "pct_at_floor")], ))

```

Biological sidebar — What do “stable” miRNAs mean in blood? miRNAs with very low CV across all disease groups (AD + MCI + Control) are behaving like biological housekeeping genes in blood. If a miRNA has high and stable expression regardless of AD status, it may reflect constitutive biological processes in blood cells (cell maintenance, basic metabolic regulation) rather than disease-specific signals. These stable miRNAs are candidates for use as reference normalization controls — analogous to the role of GAPDH in RT-qPCR. The miRNAs with high CV, on the other hand, are the ones worth investigating: their variation *might* be disease-related, though it could also be noise.

3.1.3 Zeros, Floors and Detection

How “not detected” looks depends on the platform. **RNA-seq count data** (such as GSE46579) — even after VST transformation — frequently shows **zero inflation**: a higher proportion of zero values than would be expected from a simple Gaussian or negative binomial distribution. This arises for two distinct reasons:

- 1. Structural zeros:** The miRNA is genuinely not expressed in a given sample. For example, a neuron-enriched miRNA (like miR-9-5p) may be undetectable in serum from some healthy individuals but detectable in AD patients whose neuronal EV release is increased.
- 2. Sampling zeros:** The miRNA is expressed at very low levels, but by chance no reads mapping to it were captured in that library. This is a technical artifact.

Distinguishing these two types is important: structural zeros carry biological information (the miRNA is truly absent in that sample), while sampling zeros are noise. Unfortunately, they are mathematically indistinguishable in a single experiment.

Microarray data (such as GSE120584) has no zeros: a miRNA below background is reported at the sample's floor value. The same two situations exist — truly absent vs present but below the detection limit — and are equally indistinguishable. So on an array we count **samples at the detection floor** instead of zeros:

```
# On RNA-seq you would count zeros; on this microarray there are none
cat("Zero values in the matrix:", sum(expr == 0), "(a microarray never reports 0)\n")

# The microarray equivalent: how often is each miRNA at the detection floor?
cat("miRNAs at the floor in > 20% of samples:", sum(mirna_stats$pct_at_floor > 20), "\n")
cat("miRNAs at the floor in > 50% of samples:", sum(mirna_stats$pct_at_floor > 50), "\n")

library(ggplot2)
ggplot(mirna_stats, aes(x = pct_at_floor)) +
  geom_histogram(bins = 40, fill = "#4575B4", colour = "white", alpha = 0.8) +
  geom_vline(xintercept = 20, colour = "red", linetype = "dashed") +
  geom_vline(xintercept = 50, colour = "darkred", linetype = "dashed") +
  labs(
    title = "How Often Is Each miRNA Not Detected?",
    x = "% Samples at the Detection Floor (not detected)",
    y = "Number of miRNAs",
    caption = "Red dashed lines: 20% and 50% thresholds"
  ) +
  theme_bw(base_size = 12)
```

On GSE120584 the matrix contains **0 zeros**, **124 of 925 miRNAs** sit at the floor in more than 20% of samples, and **none** in more than 50%. 268 miRNAs are never at the floor, and 396 are at it in fewer than 1% of samples. Week 2's detection filter (detected in $\geq 80\%$ of the samples of at least one group) guarantees the 50% line is not crossed.

Practical decision: miRNAs with $> 50\%$ zeros (RNA-seq) or $> 50\%$ of samples at the detection floor (microarray) rarely contribute useful signal to ML models. The Week 2 filters (filterByExpr for counts; the detection-rate filter for the array) should have removed most of these, but it is worth verifying here. If $> 10\%$ of your remaining miRNAs cross the 50% line, consider tightening the filtering threshold.

3.1.4 Distribution Shape: Is the Data Well-Normalized?

After normalization (log2 array intensities for GSE120584; VST for RNA-seq counts), expression values should have a similar distribution in every sample. Checking this with density plots confirms whether normalization succeeded.

```
# Density plots for three randomly selected samples (one per group)
set.seed(42)
idx_ctrl <- sample(which(meta$group == "Control"), 1)
idx_mci <- sample(which(meta$group == "Mild Cognitive Impairment"), 1)
idx_ad <- sample(which(meta$group == "Alzheimer's Disease"), 1)

# Build a long-format data frame for ggplot
density_df <- data.frame(
  expression = c(expr[, idx_ctrl], expr[, idx_mci], expr[, idx_ad]),
  group = rep(c("Control", "Mild Cognitive Impairment", "Alzheimer's Disease"),
    each = nrow(expr))
)
```

```

)

ggplot(density_df, aes(x = expression, colour = group)) +
  geom_density(linewidth = 1.2) +
  scale_colour_manual(values = GROUP_COLOURS) +
  labs(
    title = "Expression Value Distribution (Three Representative Samples)",
    x = "Normalised log2 expression",
    y = "Density",
    colour = "Group"
  ) +
  theme_bw(base_size = 12)

```

What to look for: - All three curves should have similar overall shape (for GSE120584, a peak near the detection floor plus a long tail of detected miRNAs) - Curves shifted dramatically relative to each other suggest normalization is incomplete - A bimodal distribution (two humps) can indicate: two populations of samples mixed together, or a subset of miRNAs with very low expression pulling down one tail

MODULE 3.2 — Dimensionality Reduction: PCA in Depth

3.2.1 The Problem PCA Solves

Our variance-filtered miRNA expression matrix has 694 miRNAs and 1,328 samples. Visualizing the data directly would require a 694-dimensional plot — impossible for the human visual system. We need a way to **reduce** those 300 dimensions into 2 or 3 that capture the most important structure in the data.

Principal Component Analysis (PCA) does exactly this — and it does so in a mathematically principled way that tells us *which* combinations of miRNAs drive the largest sources of variation in the dataset.

3.2.2 Mathematical Intuition: The Covariance Analogy for Wet-Lab Biologists

Imagine you measured two proteins in 100 serum samples: Protein A (total tau) and Protein B (phosphorylated tau / p-tau). If these two proteins behave similarly across samples — samples with high total tau tend to also have high p-tau — their measurements are **correlated**. If you plotted all 100 samples as points in a 2D scatter plot (Protein A on x-axis, Protein B on y-axis), the points would form an elongated ellipse along a diagonal line.

That diagonal line is the **first principal component (PC1)** of this 2-protein dataset. It is the axis of maximum variance — the direction along which samples differ the most. A single number (the coordinate of each sample along PC1) captures most of the information that was spread across two original measurements.

PCA generalizes this to hundreds of dimensions simultaneously. Each principal component is a **linear combination** of all 694 miRNAs, chosen to capture a different dimension of variation. PC1 captures the largest source of variation; PC2 captures the second largest (and is mathematically perpendicular to PC1); and so on.

The covariance matrix: Computationally, PCA starts by calculating the **covariance matrix** of the data — a 694×694 matrix where each entry measures how much two miRNAs co-vary across samples. miRNAs that increase together in AD samples (like miR-21 and miR-146a, both elevated in neuroinflammation) will have high positive covariance. miRNAs that move in opposite directions

(like miR-29a, downregulated in AD, and miR-146a, upregulated) will have negative covariance. PCA finds the axes that best summarize this entire covariance structure.

Key insight for biologists: If PC1 separates AD samples from controls along a single axis, it means that a coordinated program of miRNA changes — a biological module — is the dominant source of variation in the dataset. The miRNAs that contribute most to PC1 (the “loadings”) are the miRNAs most responsible for separating the groups. These are your first-pass candidate biomarkers — before any formal differential expression analysis.

3.2.3 Running PCA in R

```
library(ggplot2)
library(ggrepl)

# =====
# PCA – prcomp() function
# =====
# Important: prcomp expects samples as ROWS and features as COLUMNS.
# Our expression matrix has miRNAs as rows and samples as columns.
# So we transpose: t(expr) = samples x miRNAs.
#
# scale. = TRUE: standardize each miRNA to mean 0 and SD 1 before PCA.
# This prevents miRNAs with high absolute expression from dominating PCA
# simply because they have larger absolute values.

pca_result <- prcomp(t(expr), scale. = TRUE)

# Variance explained by each PC
var_explained <- (pca_result$sdev^2) / sum(pca_result$sdev^2) * 100
cumulative_var <- cumsum(var_explained)

cat("=== Variance Explained by First 10 PCs ===\n")
for (i in 1:10) {
  cat(sprintf("PC%d: %5.1f%% (cumulative: %5.1f%%)\n",
              i, var_explained[i], cumulative_var[i]))
}
```

3.2.4 The Scree Plot: How Many PCs Matter?

A **scree plot** shows the variance explained by each principal component. Its name comes from geology: “scree” is the loose rock debris at the base of a cliff, and the plot often shows a steep drop followed by a flat scree of small components.

```
# ---- Scree plot ----
scree_df <- data.frame(
  PC = factor(1:20, levels = 1:20),
  variance = var_explained[1:20],
  cumulative = cumulative_var[1:20]
)

p_scree <- ggplot(scree_df, aes(x = PC, y = variance)) +
  geom_col(fill = "#4575B4", width = 0.7, alpha = 0.85) +
  geom_line(aes(y = cumulative, group = 1), colour = "#D73027",
```

```

    linewidth = 1, linetype = "solid") +
geom_point(aes(y = cumulative), colour = "#D73027", size = 2.5) +
geom_hline(yintercept = 80, colour = "grey40", linetype = "dashed") +
labs(
  title = "Scree Plot – PCA on GSE120584 miRNA Expression",
  x = "Principal Component",
  y = "% Variance Explained",
  caption = "Red line: cumulative variance. Dashed line: 80% cumulative threshold."
) +
theme_bw(base_size = 12) +
theme(plot.title = element_text(face = "bold"))

print(p_screes)
ggsave("results/pca_screes.png", p_screes, width = 8, height = 5, dpi = 150)

```

How to interpret the scree plot: - A steep drop from PC1 to PC2 means one dominant source of variation exists (ideal: this is disease) - A gradual decline means variation is spread across many PCs — the dataset has complex structure with multiple sources of variation (disease, age, sex, batch, other biology) - The “elbow” in the curve (where the steep drop becomes flat) guides how many PCs to include in downstream analyses - The cumulative variance line tells you: to retain 80% of total data variance, you need the first N PCs

Biological interpretation sidebar: In a well-preprocessed AD miRNA dataset, you might expect PC1 to explain 15–30% of total variance and to correlate strongly with disease group (Control vs AD). PC2 and PC3 might capture age-related variation, sex differences, or miRNA co-expression modules tied to immune cell composition. If PC1 explains >50% of variance and does not correspond to any known biological variable, this suggests residual batch effects or a strongly dominant technical artifact that ComBat did not fully remove.

What GSE120584 shows: PC1 explains 20.1% and PC2 9.5% of the variance; Controls sit on one side of PC1 and AD/MCI on the other, with heavy overlap. But PC1 correlates **0.96** with each array’s average expression — it is essentially an “overall signal level” axis. Controls have lower overall signal than AD/MCI samples, which could be biological (more circulating miRNAs in AD) or technical (sample handling, storage time, array processing; Week 2 showed Controls detect fewer miRNAs). Always check what a PC tracks before calling it a disease axis: `cor(pca_result$x[, 1], colMeans(expr_vf))`.

3.2.5 PC1/PC2 Scatter Plot: The Core Visualization

```

# ---- Build PCA data frame for plotting ----
pca_df <- data.frame(
  PC1 = pca_result$x[, 1],
  PC2 = pca_result$x[, 2],
  PC3 = pca_result$x[, 3],
  PC4 = pca_result$x[, 4],
  Group = meta$group,
  Sex = meta$sex,
  Age = meta$age,
  row.names = rownames(pca_result$x)
)

# ---- PC1 vs PC2 coloured by disease group ----
p_pc12 <- ggplot(pca_df, aes(x = PC1, y = PC2,
                             colour = Group, shape = Group)) +
  geom_point(size = 3.5, alpha = 0.85) +

```

```

scale_colour_manual(values = GROUP_COLOURS) +
scale_shape_manual(values = c(16, 17, 15)) + # circle, triangle, square
stat_ellipse(aes(group = Group, colour = Group),
             type = "norm", level = 0.80, linetype = "dashed",
             linewidth = 0.8) +

labs(
  title = "PCA: PC1 vs PC2 – Coloured by Disease Group",
  x      = paste0("PC1 (", round(var_explained[1], 1), "% variance)"),
  y      = paste0("PC2 (", round(var_explained[2], 1), "% variance)"),
  colour = "Group", shape = "Group"
) +
theme_bw(base_size = 12) +
theme(plot.title = element_text(face = "bold"),
      legend.position = "right")

print(p_pc12)
ggsave("results/pca_pc1_pc2_group.png", p_pc12, width = 8, height = 6, dpi = 150)

```

What to look for in this plot: - AD samples shifted to one end of PC1 and control samples at the other → PC1 represents a disease axis — excellent - MCI samples positioned between AD and control → the miRNA signature tracks disease progression — biologically compelling - Groups completely overlapping → no linear separation; harder ML task, may benefit from nonlinear classifiers - Outlier points far from their group's cluster → possible sample swaps, extreme cases, or misdiagnosed samples

3.2.6 Biplot: Which miRNAs Drive PC1?

A **biplot** overlays both the sample scores and the miRNA loadings on the same PCA axes. Arrows represent miRNAs; their direction and length indicate how much each miRNA contributes to each PC and in which direction.

```

library(ggrepel)

# Extract loadings for top contributing miRNAs on PC1 and PC2
loadings <- as.data.frame(pca_result$rotation[, 1:2])
loadings$mirna <- rownames(loadings)
loadings$importance <- sqrt(loadings$PC1^2 + loadings$PC2^2)

# Keep top 20 most important miRNAs for biplot clarity
top_loadings <- loadings[order(loadings$importance, decreasing = TRUE)[1:20], ]

# Scale loadings to sample score range for plotting
scale_factor <- max(abs(c(pca_df$PC1, pca_df$PC2))) /
  max(abs(c(top_loadings$PC1, top_loadings$PC2))) * 0.6

p_biplot <- ggplot(pca_df, aes(x = PC1, y = PC2, colour = Group)) +
  geom_point(size = 2, alpha = 0.6) +
  scale_colour_manual(values = GROUP_COLOURS) +
  geom_segment(data = top_loadings,
              aes(x = 0, y = 0,
                  xend = PC1 * scale_factor,
                  yend = PC2 * scale_factor),
              arrow = arrow(length = unit(0.2, "cm"), type = "closed"),
              colour = "grey30", linewidth = 0.5, inherit.aes = FALSE) +
  geom_text_repel(data = top_loadings,

```

```

aes(x = PC1 * scale_factor,
     y = PC2 * scale_factor,
     label = mirna),
size = 2.8, colour = "grey20",
max.overlaps = 20, inherit.aes = FALSE) +
labs(
  title = "PCA Biplot – Top 20 miRNA Loadings",
  x = paste0("PC1 (", round(var_explained[1], 1), "% variance)"),
  y = paste0("PC2 (", round(var_explained[2], 1), "% variance)")
) +
theme_bw(base_size = 11) +
theme(plot.title = element_text(face = "bold"))

print(p_biplot)
ggsave("results/pca_biplot.png", p_biplot, width = 9, height = 7, dpi = 150)

```

3.2.7 Loadings Table: The miRNA Contributions to PC1

The loadings table is often more informative than the biplot for identifying specific candidate miRNAs. A positive loading on PC1 means the miRNA increases along PC1; if AD samples are at the positive end of PC1, this miRNA is upregulated in AD.

```

# Detailed loadings table for PC1
pc1_loadings <- data.frame(
  mirna = rownames(pca_result$rotation),
  loading = pca_result$rotation[, 1],
  stringsAsFactors = FALSE
)
pc1_loadings <- pc1_loadings[order(abs(pc1_loadings$loading), decreasing = TRUE), ]

cat("=== Top 20 miRNA Loadings on PC1 ===\n")
cat("(Positive loading = higher expression in the positive-PC1 direction)\n\n")
print(head(pc1_loadings, 20))

# Save loadings table
write.csv(pc1_loadings, "results/pca_pc1_loadings.csv", row.names = FALSE)
cat("\nFull PC1 loadings saved to results/pca_pc1_loadings.csv\n")

```

Biological sidebar — Interpreting PC1 Loadings as a Disease Axis: If PC1 separates AD from control samples, then the miRNAs with the largest absolute loadings on PC1 are the miRNAs most responsible for that separation. A miRNA like miR-21-5p with a strong positive loading on PC1, combined with AD samples sitting at the positive end of PC1, implies miR-21-5p is elevated in AD. Cross-reference this with miRBase and miRTarBase: if this miRNA targets PTEN (a tumor suppressor that also plays roles in neuronal survival) and PDCD4, the biology is consistent with the reported anti-apoptotic activity of miR-21 in AD. This cross-referencing step — going from a computational loading number to a biological mechanism — is exactly what distinguishes a computational biologist from a bioinformatician who only reports p-values.

3.2.8 PC3 and PC4 Check

Checking PC3 and PC4 often reveals secondary sources of variation — confounders or biological sub-groups.

```

# PC3 vs PC4
p_pc34 <- ggplot(pca_df, aes(x = PC3, y = PC4,
                             colour = Group, shape = Group)) +
  geom_point(size = 3, alpha = 0.85) +
  scale_colour_manual(values = GROUP_COLOURS) +
  scale_shape_manual(values = c(16, 17, 15)) +
  labs(
    title = "PCA: PC3 vs PC4 – Check for Secondary Structure",
    x      = paste0("PC3 (", round(var_explained[3], 1), "% variance)"),
    y      = paste0("PC4 (", round(var_explained[4], 1), "% variance)")
  ) +
  theme_bw(base_size = 12)

# PC1 vs PC2 coloured by sex (confounder check)
p_pc12_sex <- ggplot(pca_df, aes(x = PC1, y = PC2,
                                  colour = Sex, shape = Group)) +
  geom_point(size = 3, alpha = 0.85) +
  labs(
    title = "PCA: PC1 vs PC2 – Coloured by Sex",
    x      = paste0("PC1 (", round(var_explained[1], 1), "% variance)"),
    y      = paste0("PC2 (", round(var_explained[2], 1), "% variance)")
  ) +
  theme_bw(base_size = 12)

library(gridExtra)
grid.arrange(p_pc34, p_pc12_sex, ncol = 2)
ggsave("results/pca_pc34_sex_check.png",
       arrangeGrob(p_pc34, p_pc12_sex, ncol = 2),
       width = 14, height = 6, dpi = 150)

```

MODULE 3.4 — Unsupervised Clustering

3.4.1 What Clustering Tells Us (and What It Does Not)

Unsupervised clustering asks: if we group samples based purely on their miRNA expression profiles, with no knowledge of clinical labels, do the resulting groups correspond to our known disease categories?

If they do — if a purely data-driven algorithm re-discovers the AD/Control/MCI grouping — this is strong evidence that the miRNA expression patterns genuinely differ between groups. It suggests that the AD signal is not just a collection of small effects scattered across hundreds of miRNAs, but a coherent coordinated pattern that is large enough to drive global sample clustering.

If they do not — if expression-based clusters split AD patients into two sub-groups or mix AD and MCI samples — this too is informative. It may indicate biological heterogeneity (AD sub-types), or that the clinical labels themselves are imprecise (some MCI patients may be further along the AD trajectory than their MMSE score suggests).

3.4.2 Hierarchical Clustering

Hierarchical clustering builds a tree (dendrogram) by iteratively merging the two most similar samples into a cluster, then merging clusters, until all samples are in one tree. The height at which branches merge indicates the dissimilarity between groups.

Linkage method — Ward.D2: Among the several available linkage methods (single, complete, average, Ward.D2), **Ward.D2 linkage** (Ward's minimum variance method with squared Euclidean distances) consistently performs best for gene expression data clustering. It minimizes the total within-cluster variance at each merge step, producing compact, well-separated clusters. This is why pheatmap uses Ward.D2 by default.

Distance metric: We use **Euclidean distance** between samples. In the normalized log2 space, Euclidean distance is appropriate because expression differences are on a comparable scale across miRNAs (after scaling). Alternative: 1 minus Pearson correlation (correlation distance) is sometimes used when relative patterns matter more than absolute levels.

```
# =====  
# Hierarchical clustering – hclust with Ward.D2  
# =====  
library(cluster)  
library(factoextra)  
  
# Compute Euclidean distance matrix between samples  
# t(expr): samples as rows, miRNAs as columns  
dist_matrix <- dist(t(expr), method = "euclidean")  
  
# Perform hierarchical clustering  
hc <- hclust(dist_matrix, method = "ward.D2")  
  
# Basic dendrogram plot  
par(mar = c(5, 4, 4, 2))  
plot(hc,  
      main = "Hierarchical Clustering Dendrogram – Ward.D2",  
      labels = meta$group,  
      cex = 0.5,  
      xlab = "Samples",  
      ylab = "Height (Ward.D2 dissimilarity)")  
  
# Add group-coloured rectangle at k = 3  
rect.hclust(hc, k = 3, border = c("#4575B4", "#FEE090", "#D73027"))
```

Biological sidebar — Reading a Dendrogram: In a hierarchical clustering dendrogram, samples that merge at a **low height** are very similar to each other. Samples that only merge at a high height (near the top of the tree) are very different. If the dendrogram naturally separates into three branches at height h , and those three branches correspond to your three clinical groups (Control, MCI, AD), this is strong evidence that the miRNA expression signature tracks disease stage. When you see AD and MCI samples interleaved in the dendrogram — mixing between branches — consider two biological interpretations: (1) some MCI patients are at an advanced stage similar to AD, or (2) some AD patients have a relatively mild molecular phenotype. Both are biologically plausible and worth noting in your results.

3.4.3 Determining Optimal Number of Clusters: Gap Statistic and Silhouette

The number of clusters k is not given to us by the algorithm — we must choose it. Two complementary methods help:

Gap statistic (Tibshirani et al., 2001): Compares the observed within-cluster variation with that expected under a null distribution of no clustering. The optimal k is where the gap statistic is highest, or where adding another cluster stops improving the gap by a meaningful amount. The gap statistic accounts for both the quality of clustering and the expected behavior by chance — it

penalizes overfitting.

Silhouette width: For each sample, the silhouette measures how similar it is to its own cluster compared to the nearest neighboring cluster. Values range from -1 (wrong cluster) to +1 (perfectly assigned). The average silhouette width across all samples reaches a maximum at the optimal k.

```
# ---- Gap statistic ----
set.seed(42)
gap_stat <- clusGap(
  t(expr),
  FUN      = hcut,          # hierarchical clustering function
  K.max     = 8,            # try up to 8 clusters
  B         = 50,          # number of bootstrap resamples (increase to 500 for publication)
  nstart    = 25
)

p_gap <- fviz_gap_stat(gap_stat) +
  labs(title = "Gap Statistic for Optimal Number of Clusters") +
  theme_bw(base_size = 12)
print(p_gap)
ggsave("results/gap_statistic.png", p_gap, width = 7, height = 5, dpi = 150)

cat("Optimal k by gap statistic:", which.max(gap_stat$Tab[, "gap"]), "\n")

# ---- Silhouette width ----
sil_widths <- numeric(7)
for (k in 2:8) {
  clusters <- cutree(hc, k = k)
  sil      <- silhouette(clusters, dist_matrix)
  sil_widths[k - 1] <- mean(sil[, "sil_width"])
}

sil_df <- data.frame(k = 2:8, sil_width = sil_widths)

p_sil <- ggplot(sil_df, aes(x = k, y = sil_width)) +
  geom_line(colour = "#D73027", linewidth = 1.2) +
  geom_point(size = 3.5, colour = "#D73027") +
  geom_vline(xintercept = sil_df$k[which.max(sil_df$sil_width)],
    linetype = "dashed", colour = "grey40") +
  labs(
    title = "Silhouette Width vs Number of Clusters",
    x      = "Number of Clusters (k)",
    y      = "Average Silhouette Width"
  ) +
  theme_bw(base_size = 12)
print(p_sil)
ggsave("results/silhouette_width.png", p_sil, width = 7, height = 5, dpi = 150)

cat("Optimal k by silhouette:", sil_df$k[which.max(sil_df$sil_width)], "\n")
```

3.4.4 Cluster Purity Analysis

Once we have chosen k and cut the dendrogram, we evaluate how well the expression-based clusters match the known clinical labels — this is **cluster purity**.

```

# Cut dendrogram at k = 3 (assuming this is optimal)
k_optimal <- 3
cluster_labels <- cutree(hc, k = k_optimal)

# Cross-tabulation: expression clusters vs clinical groups
purity_table <- table(
  Cluster      = factor(cluster_labels, labels = paste0("Cluster_", 1:k_optimal)),
  Clinical_Group = meta$group
)

cat("=== Cluster Purity Table ===\n")
cat("(Rows = expression-based clusters; Columns = known clinical labels)\n\n")
print(purity_table)

# Compute purity per cluster (dominant group / total in cluster)
purity_per_cluster <- apply(purity_table, 1, function(row) max(row) / sum(row))
overall_purity      <- sum(apply(purity_table, 1, max)) / sum(purity_table)

cat("\nPurity per cluster:\n")
print(round(purity_per_cluster, 3))
cat("\nOverall cluster purity:", round(overall_purity, 3), "\n")
cat("(1.0 = perfect; > 0.70 = acceptable for unsupervised discovery)\n")

write.csv(as.data.frame.matrix(purity_table),
  "results/cluster_purity_table.csv")

```

Biological interpretation: An overall cluster purity of 0.75 means that 75% of samples are in the cluster dominated by their known clinical group. For an unsupervised analysis — which uses no label information — this is a strong result. It means the miRNA expression patterns are coherent enough within clinical groups that a label-blind algorithm recovers roughly the right groupings.

What is the chance baseline? Purity is never 0, even when the clusters are meaningless. If every sample were placed in a single cluster, purity would equal the proportion of the largest clinical group — about 0.33 for three equal-sized groups, but **1,009 / 1,328 = 0.76 for GSE120584**, because AD makes up 76% of the cohort. Random cluster assignments give a value close to this baseline (slightly higher with more clusters). Always judge purity against this baseline, not against 0 or 0.50: a purity near the largest-group proportion suggests the miRNA profiles do not differ systematically between groups, which is concerning for the downstream ML analysis. With three equal groups, a purity of 0.68 would be about twice the chance level.

What GSE120584 shows: hierarchical clustering at $k = 3$ gives an overall purity of **0.76 — exactly the chance level** — because every cluster is majority AD (69%, 89% and 70%). Silhouette widths are near zero (≈ 0.06 at the best $k = 2$), and the gap statistic keeps rising up to the largest k tested. The serum miRNA profiles do not form distinct groups that match the diagnosis; any disease signal is subtle and spread over many miRNAs. With unbalanced groups, prefer chance-corrected metrics such as the adjusted Rand index.

Caution: purity always increases with the number of clusters (with $k =$ number of samples it reaches 1.0), so compare purity values only at the same k .

3.4.5 k-means Clustering

k-means clustering assigns each sample to one of k clusters by iteratively minimizing the sum of squared distances from each sample to its cluster centroid. Unlike hierarchical clustering, k-means does not produce a dendrogram; it directly assigns cluster memberships.

k-means is faster and scales better to large datasets than hierarchical clustering. It is also more sensitive to the initial random placement of centroids, which is why running it multiple times (nstart) with different starting conditions is essential.

```
# ---- k-means with k=3 ----
set.seed(42)
km <- kmeans(t(expr), centers = 3, nstart = 50, iter.max = 200)

# Cross-tabulation with clinical groups
km_purity <- table(
  kmeans_cluster = factor(km$cluster, labels = paste0("kCluster_", 1:3)),
  Clinical_Group = meta$group
)
cat("=== k-means Cluster Purity ===\n")
print(km_purity)

km_overall_purity <- sum(apply(km_purity, 1, max)) / sum(km_purity)
cat("\nk-means overall purity:", round(km_overall_purity, 3), "\n")

# Visualize k-means result on PCA
pca_df$kmeans_cluster <- factor(km$cluster)

p_km_pca <- ggplot(pca_df, aes(x = PC1, y = PC2,
                              colour = Group, shape = kmeans_cluster)) +
  geom_point(size = 3.5, alpha = 0.85) +
  scale_colour_manual(values = GROUP_COLOURS) +
  labs(
    title = "k-means Clustering (k=3) Overlaid on PCA",
    x = paste0("PC1 (", round(var_explained[1], 1), "% variance)"),
    y = paste0("PC2 (", round(var_explained[2], 1), "% variance)"),
    colour = "Clinical Group", shape = "k-means Cluster"
  ) +
  theme_bw(base_size = 12)
print(p_km_pca)
ggsave("results/kmeans_pca_overlay.png", p_km_pca, width = 8, height = 6, dpi = 150)
```

MODULE 3.5 — Heatmap Visualization

3.5.1 The Heatmap as a Biological Reading Tool

A heatmap is a matrix where cells are colored by value — red for high expression, blue for low, white for average — and rows and columns are reordered by hierarchical clustering so that similar miRNAs and similar samples are adjacent. For miRNA expression data in AD research, a well-constructed heatmap is often the single most informative visualization in a paper.

Reading a heatmap requires understanding three layers simultaneously: - **The color of each cell:** How is this miRNA expressed in this sample relative to average? - **The row clustering:** miRNAs clustered together have correlated expression patterns — they may be co-regulated or involved in the same biological pathway - **The column clustering:** Samples clustered together have similar overall miRNA profiles — they may belong to the same disease group, be the same sex, or have

another shared characteristic

3.5.2 Building a Publication-Quality Heatmap in R

```
library(pheatmap)
library(RColorBrewer)

# =====
# Select top 50 most variable miRNAs for heatmap
# =====
# Using IQR (robust to outliers) rather than SD
iqr_per_mirna <- apply(expr, 1, IQR)
top50_mirnas <- names(sort(iqr_per_mirna, decreasing = TRUE))[1:50]
expr_top50 <- expr[top50_mirnas, ]

# =====
# Annotation tracks (sidebar bars on top of heatmap columns)
# =====
# Prepare annotation data frame (one row per sample, matching column order)
annotation_col <- data.frame(
  Group = meta$group,
  row.names = colnames(expr_top50)
)

if ("sex" %in% colnames(meta)) {
  # One spelling per sex, whatever the dataset used ("Male", "male", "M" → "male")
  sex_first_letter <- tolower(substr(meta$sex, 1, 1))
  annotation_col$Sex <- factor(c(f = "female", m = "male")[sex_first_letter],
    levels = c("female", "male"))
}

if ("age" %in% colnames(meta) && !all(is.na(meta$age))) {
  annotation_col$Age_Group <- cut(
    meta$age,
    breaks = c(59, 69, 74, 79, 100),
    labels = c("60-69", "70-74", "75-79", "80+"),
    include.lowest = TRUE
  )
}

# Define colours for annotation tracks
ann_colours <- list(
  Group = GROUP_COLOURS,
  Sex = c("female" = "#B2182B", "male" = "#2166AC"),
  Age_Group = c("60-69" = "#FFF3FF", "70-74" = "#BDD7E7",
    "75-79" = "#6BAED6", "80+" = "#2171B5")
)
ann_colours <- ann_colours[intersect(names(ann_colours), colnames(annotation_col))]

# =====
# Generate the heatmap
# =====
pheatmap(
  expr_top50,
```

```

scale           = "row",           # z-score each miRNA across samples
color           = colorRampPalette(rev(brewer.pal(11, "RdBu")))(100),
breaks         = seq(-3, 3, length.out = 101), # clamp at +/- 3 SD
clustering_method = "ward.D2",
clustering_distance_rows = "euclidean",
clustering_distance_cols = "euclidean",
annotation_col  = annotation_col,
annotation_colors = ann_colours,
show_rownames   = TRUE,
show_colnames   = FALSE,
fontsize_row    = 7,
main           = "Top 50 Most Variable miRNAs – GSE120584\n(row-scaled, Ward.D2 clustering)",
filename       = "results/heatmap_top50_miRNAs.png",
width          = 12,
height         = 14
)
cat("Heatmap saved to results/heatmap_top50_miRNAs.png\n")

```

3.5.3 Row and Column Scaling — Why It Matters

Row scaling (subtracting each miRNA's mean and dividing by its SD across samples) is the standard approach for expression heatmaps. It ensures that the color in each cell reflects *relative* expression of that miRNA in that sample — is it higher or lower than this miRNA's average? Without row scaling, a single highly-expressed miRNA (like miR-21-5p with mean expression 12) would have all cells colored the same intense red and would visually dominate the heatmap, making it impossible to see the biological pattern in lower-expressed miRNAs.

Column scaling (scaling each sample rather than each miRNA) is rarely appropriate for expression heatmaps — it makes every sample look equally “variable,” which destroys information about samples that are genuinely more aberrant than others.

No scaling is sometimes used when you want to show absolute expression levels — for example, when comparing one particular miRNA's absolute level across clinical groups for a clinical cutoff analysis. This is not typical for exploratory heatmaps.

3.5.4 How to Read a Heatmap Biologically

Step through these interpretive questions every time you look at a heatmap:

1. Do the columns cluster by clinical group? If the column dendrogram separates AD samples on one side and control samples on the other, the miRNA expression signatures are sufficiently different to drive clustering. If samples are mixed, the signal may be too weak, or confounders are dominating.

2. What do the top row clusters look like? The top rows in a Ward.D2 clustered heatmap (those that form the largest row clusters) represent the miRNAs with the most correlated expression patterns. A cluster of 5-10 miRNAs all showing high expression in AD samples (red column block) represents a co-regulated gene module — potentially all regulated by the same transcription factor, or all released from the same cell type in AD.

3. Are there annotation track patterns? A perfect alignment between the Group annotation bar and column clusters is ideal. But watch the other tracks: if the Sex bar correlates perfectly with the column clusters (all females in one cluster, all males in another), sex may be confounding the disease signal. Similarly for Age_Group — if older patients cluster together regardless of disease status, age is a major source of variance.

4. Which miRNAs are cluster-defining? Extract the row names in each main row cluster. For each cluster, ask: are these known AD-associated miRNAs? Look up their targets in miRTarBase. A cluster containing miR-29a, miR-107, and miR-132 — all known to target BACE1 or tau kinases — is biologically coherent and strongly suggests this is a real biological signal, not a noise artifact.

Specific miRNAs to look for in top cluster-defining rows: - **miR-29 family (miR-29a-3p, miR-29b-3p, miR-29c-3p):** All target BACE1; expect downregulation in AD cluster rows - **miR-132-3p / miR-212-3p:** Neuroprotective; target tau kinases CDK5 and GSK3-beta; expect downregulation in AD - **miR-146a-5p:** Neuroinflammation master regulator via NF-kB; expect upregulation in AD - **miR-21-5p:** Anti-apoptotic; targets PTEN; expect upregulation in AD - **miR-107:** Targets BACE1 and Cofilin; one of earliest changed miRNAs; expect downregulation. Seeing these specific miRNAs dominating the row clusters is a positive signal that the computational results align with the published biology — an important sanity check before building ML models.

MODULE 3.6 — Identifying Confounders and Outliers

3.6.1 What Is a Confounder?

A **confounder** is a variable that is associated with both the exposure (disease group) and the outcome (miRNA expression), potentially creating a spurious or inflated association. In a blood miRNA AD study, the primary confounders are:

- **Age:** AD patients are older on average than controls; age independently affects miRNA expression in blood. If miR-34a increases with age, and AD patients are older than controls, a finding of “miR-34a upregulated in AD” might simply reflect age differences, not disease.
- **Sex:** AD disproportionately affects women (approximately 2:1 female:male), and sex influences many circulating miRNA levels. A miRNA found consistently elevated in women (regardless of disease status) could appear “AD-elevated” if the cohort has more female AD patients than female controls.
- **Cell-type composition:** Blood is a mixture of immune cells (monocytes, T-cells, B-cells, NK cells, platelets). If the immune cell proportions differ between AD patients and controls — which they do, due to AD-associated chronic inflammation — then miRNA changes attributed to AD may actually reflect changes in immune cell composition.

3.6.2 Quantifying Confounder Effects on Principal Components

The correlation between each principal component and a potential confounder variable tells us how much of the variance captured by that PC is attributable to the confounder rather than disease.

```
# =====
# Correlate each PC with age and sex
# =====

# Build PC score matrix (samples x PCs)
pc_scores <- pca_result$x[, 1:10] # first 10 PCs

# Age correlation (continuous variable – Pearson correlation)
age_cor_results <- data.frame(
  PC      = paste0("PC", 1:10),
  r_age   = NA_real_,
  p_age   = NA_real_,
  stringsAsFactors = FALSE
)
```

```

if ("age" %in% colnames(meta) && !all(is.na(meta$age))) {
  for (i in 1:10) {
    ct <- cor.test(pc_scores[, i], meta$age, method = "pearson")
    age_cor_results$r_age[i] <- ct$estimate
    age_cor_results$p_age[i] <- ct$p.value
  }
}

# Sex correlation (binary variable – point-biserial correlation via cor.test)
sex_cor_results <- data.frame(
  PC      = paste0("PC", 1:10),
  r_sex   = NA_real_,
  p_sex   = NA_real_,
  stringsAsFactors = FALSE
)

if ("sex" %in% colnames(meta)) {
  sex_binary <- as.numeric(factor(meta$sex)) - 1 # 0/1 encoding
  for (i in 1:10) {
    ct <- cor.test(pc_scores[, i], sex_binary, method = "pearson")
    sex_cor_results$r_sex[i] <- ct$estimate
    sex_cor_results$p_sex[i] <- ct$p.value
  }
}

# Combine into a single summary table
confounder_table <- data.frame(
  PC      = paste0("PC", 1:10),
  Var_Exp_pct = round(var_explained[1:10], 2),
  r_age    = age_cor_results$r_age,
  p_age    = age_cor_results$p_age,
  r_sex    = sex_cor_results$r_sex,
  p_sex    = sex_cor_results$p_sex
)

cat("=== Confounder Correlation with Principal Components ===\n")
print(confounder_table)

write.csv(confounder_table, "results/pc_confounder_correlations.csv", row.names = FALSE)

```

3.6.3 Partial R²: Partitioning Variance Between Biology and Confounders

Pearson correlation tells us the direction and strength of association, but a variance decomposition tells us the *proportion* of variance in each PC explained by each variable, holding other variables constant. This allows us to say, for example: “PC2 is 35% explained by age, 8% explained by sex, and 12% explained by disease group.”

We fit a linear model $PC1 \sim \text{group} + \text{age} + \text{sex}$ and use a Type II ANOVA, which gives each variable a sum of squares (SS) measured *after* adjusting for the other variables, plus a residual SS for the unexplained variation. Two related quantities can be built from this table:

Quantity	Formula	Interpretation
Share of variance (what the script computes; also called eta-squared, η^2)	$SS_{\text{variable}} / (SS_{\text{group}} + SS_{\text{age}} + SS_{\text{sex}} + SS_{\text{residual}})$	The fraction of the PC's variance attributable to this variable. Shares of all variables plus the residual add up to 1
Partial R² (textbook definition; also called partial eta-squared)	$SS_{\text{variable}} / (SS_{\text{variable}} + SS_{\text{residual}})$	The fraction of the variance <i>left over after the other variables are accounted for</i> that this variable explains. Values do not add up to 1 across variables

The code below reports the **share of variance**. It is labelled “partial R²” in the script and output for simplicity, but the share is the more useful quantity here: it answers “how much of PC1 is age vs disease?” on a common scale. The true partial R² is always at least as large as the share, and the two are close when the variable explains only a small part of the PC. When predictors are correlated (e.g., AD patients are older), Type II sums of squares do not add up exactly to the total, so treat the shares as approximate.

```
# Partial R2 using linear regression
# Model: PC1 ~ group + age + sex

if (all(c("age", "sex") %in% colnames(meta))) {

  model_df <- data.frame(
    PC1 = pc_scores[, 1],
    PC2 = pc_scores[, 2],
    group = meta$group,
    age = meta$age,
    sex = meta$sex
  )
  model_df <- na.omit(model_df)

  fit_pc1 <- lm(PC1 ~ group + age + sex, data = model_df)
  fit_pc2 <- lm(PC2 ~ group + age + sex, data = model_df)

  library(car)
  anova_pc1 <- Anova(fit_pc1, type = "II")
  anova_pc2 <- Anova(fit_pc2, type = "II")

  ss_total_pc1 <- sum(anova_pc1$"Sum Sq")
  partial_r2_pc1 <- anova_pc1$"Sum Sq" / ss_total_pc1

  ss_total_pc2 <- sum(anova_pc2$"Sum Sq")
  partial_r2_pc2 <- anova_pc2$"Sum Sq" / ss_total_pc2

  cat("=== Partial R-squared for PC1 ===\n")
  cat(sprintf(" Disease group: %.3f (%.1f%%)\n",
    partial_r2_pc1[1], partial_r2_pc1[1] * 100))
  cat(sprintf(" Age: %.3f (%.1f%%)\n",
    partial_r2_pc1[2], partial_r2_pc1[2] * 100))
  cat(sprintf(" Sex: %.3f (%.1f%%)\n\n",
    partial_r2_pc1[3], partial_r2_pc1[3] * 100))

  cat("=== Partial R-squared for PC2 ===\n")
  cat(sprintf(" Disease group: %.3f (%.1f%%)\n",
    partial_r2_pc2[1], partial_r2_pc2[1] * 100))
  cat(sprintf(" Age: %.3f (%.1f%%)\n",
```

```

        partial_r2_pc2[2], partial_r2_pc2[2] * 100))
cat(sprintf(" Sex:           %.3f (%.1f%%)\n",
        partial_r2_pc2[3], partial_r2_pc2[3] * 100))
}

```

Note that `partial_r2_pc1` has four entries: the last one (Residuals) is the share of PC1 variance that group, age and sex together leave unexplained. To compute the textbook partial R^2 as well, divide each variable's SS by that SS plus the residual SS:

```

# Textbook partial R^2 = SS_variable / (SS_variable + SS_residual)
ss_pc1      <- anova_pc1$"Sum Sq"
ss_resid_pc1 <- ss_pc1[length(ss_pc1)] # last row = Residuals
true_partial_r2 <- ss_pc1[-length(ss_pc1)] / (ss_pc1[-length(ss_pc1)] + ss_resid_pc1)
names(true_partial_r2) <- rownames(anova_pc1)[-nrow(anova_pc1)]
round(true_partial_r2, 3)

```

Biological sidebar — How to respond to confounder findings: If the partial R^2 analysis reveals that age explains 25% of PC1 and disease group explains only 15%, this is a serious finding with direct consequences for Week 4 modeling. It means that a significant portion of the miRNA variance your ML model will be trained on is driven by age differences, not disease biology. The appropriate response is: include age as a covariate in the Week 4 model (for GSE120584's limma analysis: `model.matrix(~ 0 + group + sex + age)`; for GSE46579's DESeq2 analysis: `~ sex + age + group`), consider matching AD patients with controls on age distribution, and report this finding explicitly in your paper's methods section. This is not a failure of the study — confounders are present in virtually every real clinical dataset. Acknowledging and accounting for them is the mark of rigorous science.

What GSE120584 shows: the shares are small — disease group 3.4%, age 0.6% and sex 0.1% of PC1; 0.2%, 1.8% and 0.5% of PC2 — because the top PCs are dominated by other variation (PC1 by overall array signal). Small values on the top PCs do **not** mean age can be ignored: age differs strongly between groups (AD 79 vs Control 72 years) and correlates with several PCs ($r \approx \pm 0.17$, $p < 10^{-9}$). Week 4 therefore includes both age and sex as covariates.

3.6.4 Identifying Sample Outliers

Beyond formal QC metrics (NUSE, library size), EDA sometimes reveals outlier samples that passed Week 2 QC but are biologically anomalous — extreme samples in PCA space that sit far from all other samples in their group.

```

# =====
# Identify outlier samples in PCA space using Mahalanobis distance
# =====
library(MASS)

identify_pca_outliers <- function(pca_df_input, group_col, threshold_pct = 97.5) {
  outlier_flags <- rep(FALSE, nrow(pca_df_input))
  groups       <- unique(pca_df_input[[group_col]])

  for (g in groups) {
    idx    <- which(pca_df_input[[group_col]] == g)
    coords <- as.matrix(pca_df_input[idx, c("PC1", "PC2")])
    if (nrow(coords) < 4) next

    tryCatch({
      cov_matrix <- cov(coords)

```

```

    centroid <- colMeans(coords)
    mah_dist <- mahalanobis(coords, centroid, cov_matrix)
    threshold <- qchisq(threshold_pct / 100, df = 2)
    outlier_flags[idx[mah_dist > threshold]] <- TRUE
  }, error = function(e) NULL)
}
outlier_flags
}

pca_df$is_outlier <- identify_pca_outliers(pca_df, "Group")

cat("Potential outlier samples:\n")
print(rownames(pca_df)[pca_df$is_outlier])

# Visualize outliers on PCA
p_outliers <- ggplot(pca_df, aes(x = PC1, y = PC2, colour = Group,
                                shape = is_outlier)) +
  geom_point(aes(size = is_outlier), alpha = 0.85) +
  scale_colour_manual(values = GROUP_COLOURS) +
  scale_shape_manual(values = c(16, 4)) +
  scale_size_manual(values = c(2.5, 5)) +
  geom_text_repel(data = pca_df[pca_df$is_outlier, ],
                 aes(label = rownames(pca_df)[pca_df$is_outlier]),
                 size = 3, colour = "black") +
  labs(
    title = "PCA – Mahalanobis Outlier Detection",
    x = paste0("PC1 (", round(var_explained[1], 1), "% variance)"),
    y = paste0("PC2 (", round(var_explained[2], 1), "% variance)"),
    colour = "Clinical Group"
  ) +
  theme_bw(base_size = 12)
print(p_outliers)
ggsave("results/pca_outlier_detection.png", p_outliers, width = 8, height = 6, dpi = 150)

# BIOLOGICAL DECISION RULE:
# An extreme AD sample with all known AD miRNA signatures elevated is NOT an error –
# it may be a severe case worth retaining.
# An outlier sample with a completely inverted signature (AD sample with control-like
# profile) warrants individual review: check metadata, re-examine clinical diagnosis,
# check for labeling errors before any exclusion decision.

```

WEEK 3 LAB SESSIONS

Lab 3A — R-Based EDA (90 minutes)

Objective: Generate a complete EDA report for GSE120584 in R, covering descriptive statistics, PCA, hierarchical clustering, and a publication-quality heatmap.

Setup:

```

# Load clean data from Week 2 (confirm this file exists)
expr <- readRDS("data/processed/GSE120584_expr_clean.rds")
meta <- readRDS("data/processed/GSE120584_metadata_clean.rds")
cat("Expression matrix:", nrow(expr), "miRNAs x", ncol(expr), "samples\n")
cat("Groups:\n")

```

```
print(table(meta$group))
```

Tasks (complete in order):

1. Descriptive statistics table (20 min):

- Run the per-miRNA statistics code from Module 3.1.2
- Identify the top 5 most variable miRNAs (by CV) and look each one up in miRBase — what does each one do biologically?
- How many miRNAs have >30% zeros? Why is the answer 0 for this microarray? How many miRNAs sit at the detection floor in >30% of samples instead?

2. PCA and scree plot (25 min):

- Run `prcomp()` and generate the scree plot
- Generate the PC1/PC2 scatter colored by group
- How much variance does PC1 explain? Does it separate the three groups?
- Print the top 10 PC1 loadings. Look up the number 1 loading miRNA in miRTarBase and identify its top 3 validated targets. Are any of these targets relevant to AD pathology?

3. Hierarchical clustering (20 min):

- Run `hclust()` with Ward.D2 linkage
- Run gap statistic and silhouette analysis (B=20 is sufficient for the lab session)
- Cut the tree at k=3 and compute cluster purity
- Does the purity suggest the expression profiles are separating by clinical group?

4. Heatmap (25 min):

- Run the heatmap code from Module 3.5.2
- Examine the output: identify one row cluster that shows high expression in the AD column block and low expression in the Control block
- List the miRNA names in that cluster and annotate them with their known AD biology

Deliverable: A brief written interpretation (1 paragraph per analysis, 4 paragraphs total) answering the biological questions raised above.

WEEK 3 ASSIGNMENTS

Reading Assignment (Required)

1. **Ringnér M (2008)** — *What is principal component analysis?* *Nature Biotechnology* 26(3):303-304. DOI: [10.1038/nbt0308-303](https://doi.org/10.1038/nbt0308-303)
Focus on: Figure 1 (the geometric interpretation of PCA); the explanation of eigenvalues and loadings. This short commentary is the clearest non-mathematical explanation of PCA in the bioinformatics literature.

Reflection Questions (Discuss in Week 4 opening)

1. You run PCA and find that PC1 explains 22% of variance and corresponds to disease group, while PC2 explains 18% and corresponds strongly to age (partial $R^2 = 0.38$ for age). What does this tell you about the relative contribution of disease biology vs aging to the miRNA expression pattern? How would you account for this in your Week 4 ML model design?
2. The cluster purity analysis returns an overall purity of 0.68. A colleague says this is “too low to be useful.” What would you say in response? What is the appropriate comparison baseline, and what does purity of 0.68 actually tell us about the dataset?

Practical Exercise

Using the heatmap generated in Lab 3A, extract the miRNA names from the largest cluster that shows high expression in AD samples and low expression in controls (a “high-in-AD” cluster). Look up these miRNAs in miRTarBase (<https://mirtarbase.cuhk.edu.cn>). Write a 3-5 sentence biological

interpretation: What known targets do these miRNAs share? What biological pathways are they likely co-regulating? Is the pattern consistent with known AD biology?

WEEK 3 GLOSSARY

Term	Definition
Principal Component (PC)	A linear combination of all features (miRNAs) that captures a specific dimension of variance; PCs are orthogonal (perpendicular) to each other and ordered by decreasing variance explained
Loading	The coefficient of each original variable (miRNA) in a principal component; a large absolute loading means that miRNA contributes strongly to that PC's variance
Scree plot	A bar chart of variance explained by each principal component in order; the "elbow" indicates the optimal number of PCs to retain
Biplot	A PCA visualization showing both sample scores and feature loadings (as arrows) on the same axes
Hierarchical clustering	An agglomerative algorithm that iteratively merges samples into a tree (dendrogram) based on pairwise distances; no k needs to be specified in advance
Ward.D2 linkage	A hierarchical clustering linkage method that minimizes within-cluster variance at each merge; produces compact, well-separated clusters; recommended for expression data
Dendrogram	The tree diagram output of hierarchical clustering; branch height represents dissimilarity between merged groups
Gap statistic	A method to determine optimal cluster number k by comparing observed within-cluster variation to that expected under a null distribution of no clustering
Silhouette width	Per-sample metric (range -1 to +1) measuring how well a sample fits its assigned cluster vs the nearest other cluster; average across samples maximizes at the optimal k
Cluster purity	The fraction of samples in each cluster that belong to the dominant clinical group; measures how well unsupervised clusters recover known labels
Cluster purity baseline	The purity obtained with no real structure: the proportion of the largest clinical group (≈ 0.33 for three equal groups; 0.76 for GSE120584). Purity must be judged against this, not against 0
Row scaling (z-score)	Subtracting a miRNA's mean and dividing by its SD across all samples before heatmap plotting; ensures color reflects relative rather than absolute expression
Coefficient of Variation (CV)	$SD / \text{mean} \times 100\%$; a scale-free measure of relative variability; allows comparison of variability between miRNAs at different expression levels
Confounder	A variable associated with both the exposure (disease group) and the outcome (miRNA expression) that can create spurious or inflated associations
Partial R-squared	Strictly, $SS_{\text{variable}} / (SS_{\text{variable}} + SS_{\text{residual}})$: the proportion of variance <i>not explained by the other predictors</i> that one predictor explains. In this course's script the label is used for the closely related share of variance ($SS_{\text{variable}} / \text{total SS}$), which partitions a PC's variance between disease, age, sex and residual

Term	Definition
Mahalanobis distance	A multivariate distance measure that accounts for correlations between dimensions; used to identify outlier samples in PCA space
Zero inflation	The phenomenon of having more zero values in a dataset than expected from the underlying statistical distribution; common in RNA-seq count data (microarrays report a detection floor instead)

KEY REFERENCES (Week 3)

All references retrieved from PubMed.

1. Jolliffe IT, Cadima J (2016). Principal component analysis: a review and recent developments. *Philos Trans R Soc A Math Phys Eng Sci* 374(2065):20150202. DOI: [10.1098/rsta.2015.0202](https://doi.org/10.1098/rsta.2015.0202)
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3. Tibshirani R, Walther G, Hastie T (2001). Estimating the number of clusters in a data set via the gap statistic. *J R Stat Soc Series B Stat Methodol* 63(2):411–423. DOI: [10.1111/1467-9868.00293](https://doi.org/10.1111/1467-9868.00293)
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7. Leek JT, Storey JD (2007). Capturing heterogeneity in gene expression studies by surrogate variable analysis. *PLoS Genet* 3(9):e161. DOI: [10.1371/journal.pgen.0030161](https://doi.org/10.1371/journal.pgen.0030161)

NEXT WEEK PREVIEW

Week 4: Feature Selection and Classical Machine Learning

Having explored the structure of the data, we are now ready to build the first supervised machine learning models for AD classification.

In Week 4 you will: - Run formal differential expression analysis (limma for the GSE120584 microarray; DESeq2 for the GSE46579 RNA-seq counts) to identify miRNAs that are statistically significantly different between AD, MCI, and Control - Understand the multiple testing problem and apply Benjamini-Hochberg FDR correction — what an adjusted p-value of 0.05 actually means when testing ~900 miRNAs simultaneously - Build your first ML classifiers: Logistic Regression, Linear SVM, and Random Forest - Evaluate model performance with ROC curves, AUC, sensitivity, and specificity - Implement cross-validation to get honest estimates of out-of-sample performance - Apply filter-based (variance, fold-change) and embedded (LASSO) feature selection to reduce the several-hundred-miRNA space to a focused biomarker panel of 10–30 miRNAs

The exploratory analyses you completed this week — particularly the PC1 loadings table and the heatmap cluster-defining miRNAs — will serve as a biological reference point for evaluating the Week 4 ML results. If your random forest identifies miR-29a and miR-146a as top features, and you already know from this week’s heatmap that these miRNAs show strong group-dependent

patterns, the biological coherence of the ML result is confirmed. If the ML identifies miRNAs you have never seen in the EDA, that is a red flag worth investigating carefully.

Data object you will take into Week 4: `data/processed/GSE120584_expr_varianceFiltered.rds` — the variance-filtered expression matrix (top 75% most variable miRNAs by IQR), ready for differential expression and ML feature engineering.